3 Project Summary

Overview Predictive algorithms, such as supervised machine learning algorithms, are often designed by optimizing some *loss function* measuring error on historical data. There is a long-standing debate within the machine learning literature around the principles guiding loss function choice, as this choice is consequential for *what statistics of our data* our algorithms are learning. Two prominent camps have emerged: one who advocates for *task-agnostic* loss functions, possibly combined with some "smart thresholding" as a post-processing measure, and another who advocates for *task-aware* loss design. Task-agnostic losses are sometimes easier to implement from an engineering perspective, but task-aware loss design, when principled, can lead to improved *task-specific* performance. However, the notion of *multicalibration* has emerged as formalization of fairness, and there are concerns that task-specific optimization might sacrifice multicalibration in settings where task-agnostic optimization implies multicalibration.

In much of this debate, the relationship between loss function choice and model complexity has been swept under the rug. Typically, theoretical work assumes the model class $\mathcal{H}$ is "sufficiently rich" to contain the optimal classifier, but we have little understanding if this benefit is even present if $\mathcal{H}$ is small, as is common in data-limited settings or for small engineering teams with limited compute resources. In general, we know that if $\mathcal{H}$ is very complex (relative to the prediction problem), then task-agnostic optimization, combined with post-processing, is just as good as task-aware optimization. However, when $\mathcal{H}$ is small, we do not know about the relative performance of task-aware vs task-agnostic optimization, especially once we move beyond settings where common theoretical assumptions cannot be tested, even empirically. Finally, we know that if $\mathcal{H}$ is sufficiently rich, fairness (via multicalibration) follows from loss optimization– either task-aware or task-agnostic. However, we again do not know how the choice of loss function affects this notion of fairness if $\mathcal{H}$ is small.

To this end, this proposal with draw from the literatures on $\mathcal{H}$-consistency, property elicitation, and multicalibration to (a) give guidance on loss function design by studying its relationship to the complexity of model class, (b) relate this to complexity of optimization by examining the complexity of the decision task in relationship to the model class, and (c) understand the implications of loss design on fairness (measured by multicalibration), especially for small models.

Intellectual Merit This proposal will contribute to and unify the literatures on multicalibration, $\mathcal{H}$-consistency, and property elicitation. The previous work on multicalibration has largely focused on binary classification problems, and dismissed the seeming impossibility of finding multicalibrated models for more complex tasks. For complex tasks like structured prediction tasks, this work will help guide practitioners in choosing their objective function to ensure the intended proxy generalizes well to the actual task while still being fair.

Broader Impacts While the foundation is foundational and theoretical in nature, its teachings will inform algorithm designers about the limitations and possibilities of of statistical fairness guarantees. While task-specific losses tend to improve (task-specific) performance, understanding if and when this improvement is disparately beneficial to one privileged group will inform our understand of how to develop more equitable algorithms for high-stakes and complex settings, such as image segmentation for computer-aided diagnostics, generative AI, and climate forecasting and modeling.